# Four reproducible finance analytics projects

A portfolio built on **Python + DuckDB** pipelines over free, adversarially-verified data sources — from a formal COVID event study to a vulnerability-adjusted vendor scorecard. Every number traceable to a primary-source filing, government record, or pinned public dataset.

PROJECT 01
### Post-COVID Sector Rotation
Event study & macro linkage across the 11 U.S. equity sectors, 2019–2025.

PROJECT 02
### Luxury Goods Comps
Market share, margins & valuation multiples across the listed global luxury houses.

PROJECT 03
### Fortune 500 Breakdowns
A multi-year panel of corporate America — profitability, efficiency, churn.

PROJECT 04
### Cloud, ERP & AI Vendor Intelligence
Segment revenue vs. security-risk exposure — finance × cybersecurity.

- Python
- DuckDB
- Parquet caching
- pandera quality gates
- Streamlit dashboards
- pytest
- GitHub Actions CI

PORTFOLIO OVERVIEW

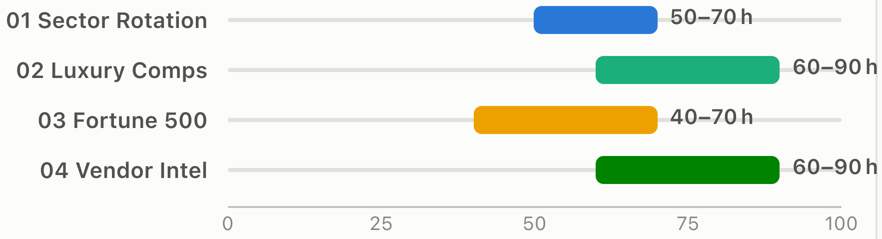
# Four projects, one engineering spine

Each project stands alone, but they share a reusable toolkit — a cached ingestion layer, a layered DuckDB warehouse, quality gates, and CI — so effort compounds across the sequence.

## THE LINEUP

| | ANGLE | DIFFICULTY | EFFORT | DATA | ROLE |
|---|---|---|---|---|---|
| 01 Sector Rotation | Event study + macro linkage | Interm.→Adv. | ~50–70 h | ★★★★★ | **Flagship** — best effort-to-impressiveness |
| 02 Luxury Comps | Comps, multiples, market share | Interm.→Adv. | ~60–90 h | ★★★☆☆ | Most **finance-recruiter** appeal |
| 03 Fortune 500 | Panel data engineering | Intermediate | ~40–70 h | ★★★★☆ | **Quick win** — builds the shared toolkit |
| 04 Vendor Intel | Finance × cybersecurity | Advanced | ~60–90 h | ★★★☆☆ | **Capstone** — most differentiated |

## ESTIMATED EFFORT (HOURS)




Range = documented estimate per project brief. Luxury and Vendor Intel run longer because PDF extraction and CVE attribution are manual-judgment work.

## RECOMMENDED BUILD ORDER

**1 · Fortune 500**
Approachable data; builds the DuckDB warehouse, SEC client & yfinance caching reused everywhere.

→

**2 · Sector Rotation**
The flagship — cleanest free data, real quant methodology (event study, CARs, t-tests).

→

**3 · Luxury Comps**
Sell-side-style comps work: IFRS/GAAP normalization, crosswalks, valuation vs. fundamentals.

→

**4 · Vendor Intel**
Advanced capstone with the highest ceiling — a rare finance × security combination.

PROJECT 01 · FLAGSHIP

# Post-COVID Sector Rotation

An event-study and macro-linkage analysis of U.S. equity sectors, 2019–2025 — how the 11 sectors crashed, recovered, and structurally re-ranked through COVID-19.

| DIFFICULTY | EFFORT | DATA CLEANLINESS | UNIVERSE | WINDOW |
|---|---|---|---|---|
| **Interm. → Adv.** | **~50–70 h** | ★★★★★ | **11 ETFs + SPY** | **2019 → 2025** |
| Real quant methodology | Best effort-to-impressiveness ratio | Fully automatable, no PDFs | XLK, XLF, XLE, XLV, XLY, XLP, XLI, XLU, XLB, XLRE, XLC | Daily prices · 8 macro series · stringency |

## THE INTERVIEW STORY

> "I ran a formal **market-model event study** of the COVID crash across all 11 GICS sectors, measured **cumulative abnormal returns vs. the S&P 500** over a 250-day estimation window, and showed which sectors were structural losers vs. winners — then linked recovery speed to sector-specific macro damage and the pandemic stringency timeline."

## THREE ANALYTICAL LAYERS

**1 · Descriptive**
Returns, rolling vol, max drawdown, recovery time, Sharpe/Sortino, correlation regimes.

**2 · Event study**
CARs vs. market-model expectation around 19 Feb → 23 Mar 2020; t-tested significance.

**3 · Macro linkage**
FRED macro + daily VIX + stringency; regress recovery speed on macro damage.

## KEY ANALYTICAL QUESTIONS

1. Which sector had the **deepest drawdown** — and does that ranking match the ranking by **recovery time**?
2. Which sectors show statistically significant negative CARs (Energy, Financials?) vs. positive (Tech, Staples?) — and does the sign **flip in the 2021–22 inflation regime**?
3. Did cross-sector correlations **spike toward 1.0** in the March-2020 panic — "diversification fails when you need it"?
4. Which sectors carried the highest **VIX beta** during the shock, and how long did volatility take to normalize?
5. Comparing 2019 vs. 2023–25 leadership: which sectors **permanently re-ranked** vs. mean-reverted?

# Sector Rotation — pipeline, data, risks

## VERIFIED DATA SOURCES

| SOURCE | ROLE | ACCESS |
|---|---|---|
| **yfinance / Stooq readers** | Primary price panel — returns, vol, drawdowns, abnormal returns | No key; fetch once, cache to Parquet |
| **FRED** | VIXCLS, UNRATE, INDPRO, CPI, sector payrolls — macro linkage & VIX beta | Free 32-char API key |
| **Our World in Data** | COVID stringency + case/death waves overlay | ⚠ Frozen 19 Aug 2024 — scope overlay to ≤2024 |
| **Oxford OxCGRT** | Containment vs. economic-support sub-indices (optional) | Direct CSV, no key |
| **BLS / BEA** | Industry payrolls; GDP-by-industry “markets lead the economy” (optional) | Free instant keys |

## FLAGSHIP VISUALS

- Drawdown vs. days-to-recover ranking
- CAR paths, significance-flagged
- Correlation-regime heatmaps
- Rolling-vol small multiples
- VIX-beta ranked bar
- 2019 vs. 2025 leadership bump chart

## METHODOLOGY

1. **Ingest once, cache to Parquet** — prices, FRED, OWID; retry/back-off.
2. Land raw in **DuckDB**; staging computes returns, aligns calendars.
3. **Descriptive marts** — vol, drawdown, recovery, Sharpe, correlations.
4. **Event study** — per-sector α, β over 250 pre-event days; CARs + t-tests.
5. **Macro linkage** — explicit resampling; VIX betas; recovery regressions.
6. pandera gates → figures → **Streamlit** → pytest + CI → findings README.

## RISKS & GOTCHAS

- **Stooq CSV sits behind a JS anti-bot wall** — use library fetchers + cache, never naive requests.
- **yfinance is fragile** (HTTP 429) — fetch once, cache, retry; never depend on it live.
- **Frequency mismatch** (daily / monthly / quarterly) is the main analytical trap — resample explicitly.
- Event-study validity needs a **clean 2019 estimation window**; frame macro results as associations, not causation.

PROJECT 02 · RECRUITER APPEAL

# Luxury Goods Market Share & Valuation

A comps model benchmarking the listed global luxury houses — LVMH, Kering, Richemont, Hermès, Burberry, Prada, Brunello Cucinelli, Estée Lauder, Tapestry, Capri — on share, mix, margins, and multiples.

| DIFFICULTY | EFFORT | DATA CLEANLINESS | UNIVERSE | WINDOW |
|---|---|---|---|---|
| **Interm. → Adv.** | **~60–90 h** | ★★★☆☆ | **Listed peer set** | **FY2015 → 2024** |
| Real sell-side analyst work | PDF extraction is the slow part | Filings PDFs need hand-mapping | IFRS + US GAAP · EUR/CHF/GBP/USD | Annual, from primary filings |

## THE INTERVIEW STORY

> "I built a comps model across the listed global luxury houses — normalizing **IFRS vs. US-GAAP** filings and four currencies, hand-mapping each house's segment and geographic taxonomy into a common schema, then benchmarking margins and **EV/EBITDA, EV/Sales and P/E vs. Damodaran** industry averages to flag who's expensive for their fundamentals. Hermès's structural margin premium shows up cleanly."

## FOUR PILLARS

**Market share & HHI**
Revenue share *within the listed peer set* — reproducible from primary filings.

**Revenue mix**
Segment & geography, hand-mapped to a common schema across houses.

**Financial performance**
3/5/10-yr CAGRs, gross/op/net margins, currency-normalized to EUR & USD.

**Valuation**
EV/Sales, EV/EBITDA, P/E vs. Damodaran; valuation-vs-fundamentals scatters.

## KEY ANALYTICAL QUESTIONS

1. How has **peer-set HHI** evolved 2015–2024 — did the 2024 downturn consolidate or fragment share?
2. Which **segment mix** proved most resilient through the 2024 contraction?
3. Geographic exposure: who is most exposed to the **China slowdown** vs. Japan/US strength?
4. Does **Hermès's structural margin premium** show cleanly vs. Damodaran's apparel average?
5. Do multiples rank houses the same way growth and margins do — **who's expensive for their fundamentals**?
6. Where do yfinance fundamentals **diverge from SEC EDGAR** ground truth for the US filers?

# Luxury Comps — pipeline, data, risks

## VERIFIED DATA SOURCES

| SOURCE | ROLE | ACCESS |
|---|---|---|
| **SEC EDGAR XBRL** | Ground-truth financials for US filers (EL, TPR, CPRI); reconciliation benchmark | No key; User-Agent header; 10 req/s |
| **Company URDs / annual reports** | **Primary segment & geographic revenue** for all houses (IFRS filers absent from EDGAR) | ⚠ PDFs; taxonomies differ → crosswalk table |
| **yfinance** | Prices, market cap, EV inputs for multiples | ⚠ Non-US fundamentals often empty — prices only |
| **Damodaran (NYU Stern)** | Industry-average margins & multiples benchmark | ⚠ URLs overwritten each January — pin file + date |
| **Bain × Altagamma** | Total-market denominator & narrative context | Free headline figures (full decks gated) |

## FLAGSHIP VISUALS

- Market-share explorer + HHI trend
- Segment-mix stacked bars
- Geographic-exposure view
- Margin ladder w/ Damodaran markers
- Valuation-vs-fundamentals scatter
- yfinance-vs-EDGAR variance panel

## METHODOLOGY

1. Ingest EDGAR XBRL, URD PDFs, yfinance prices, Damodaran — cache all raw.
2. **Hand-map each house's segment & geo taxonomy** into a common schema — a documented crosswalk, itself a defensible artifact.
3. **Currency-normalize** to EUR & USD; align March fiscal year-ends (Richemont, Burberry).
4. Marts: revenue share, HHI, CAGRs, margins, EV/Sales, EV/EBITDA, P/E.
5. Benchmark vs. Damodaran; **reconcile yfinance vs. EDGAR** and quantify variance.
6. pandera gates → multi-page Streamlit → pytest over every formula → CI.

## RISKS & GOTCHAS

- **Taxonomies differ materially** (LVMH groups vs. Kering tiers vs. Richemont maisons) — treat the crosswalk as a transparent, first-class artifact.
- **EDGAR covers only US filers** — European/Asian financials come from PDFs (manual).
- FX convention (spot vs. average) is an **analyst choice — document it**.
- Don't scrape Statista/Macrotrends; note corporate actions (Prada–Versace, Capri/Tapestry) so comps aren't distorted.

PROJECT 03 · QUICK WIN — START HERE

# Fortune 500 Financial Breakdowns

A multi-year DuckDB panel of corporate America's largest firms — sector profitability, revenue-per-employee efficiency, geographic concentration, and year-over-year list churn.

| DIFFICULTY | EFFORT | DATA CLEANLINESS | PANEL SIZE | DELIVERABLE |
|---|---|---|---|---|
| **Intermediate** | **~40–70 h** | ★★★★☆ | **~5,000 rows** | **make pipeline** |
| Best pure data-engineering showcase | Most approachable data | Community CSVs | ~10 years × 500 companies | Reproducible on a fresh clone, free data only |

### THE INTERVIEW STORY

> "I engineered a reproducible **DuckDB warehouse over ~10 years of Fortune 500 data**, then showed that tech and pharma dominate margins while retail dominates headcount, that **revenue-per-employee varies ~20× across sectors**, and that ~3–5% of the list turns over annually with churn concentrated in energy and retail."

### CORE DELIVERABLE = THE DATA ENGINEERING

- Canonical schema at company-year grain: rank, revenue, profit, assets, employees, sector, HQ + derived **net margin, revenue-per-employee, rank change**.
- **Cross-year entity resolution** — track firms through rebrands and M&A via a documented alias/override table.
- Fact/dimension model + **SQL window functions** for churn flags and rank movement.
- Quality gates: exactly 500 ranks/year, non-null keys, non-negative financials.

### KEY ANALYTICAL QUESTIONS

1. **Does size predict profitability?** Or does net margin decouple from scale, overall and by sector?
2. Which sectors are most/least profitable (median margin, ROA) — and how have rankings shifted YoY?
3. How does **revenue-per-employee** vary — which industries are the ~20× outliers?
4. How concentrated is the list (top-10/25/50 share, HHI) and **where is corporate America headquartered**?
5. **How churny is the list?** Entry/exit rates, sector churn as a disruption proxy, survival of former top-100 firms.

# Fortune 500 — pipeline, data, risks

## VERIFIED DATA SOURCES

| SOURCE | ROLE | ACCESS |
|---|---|---|
| **cmusam/fortune500 (GitHub)** | Longitudinal spine → base fact table, company-year grain | ⚠ Per-year files 2005–2019 (combined CSV 404s) — pin commit SHA, mirror raw |
| **Kaggle datasets** | Enrichment: sector, industry, HQ, employees, market cap | Free token; prefer CC0/CC BY; pin slug + version |
| **Wikipedia** | Current-year refresh & cross-check (~top 100) | pandas.read_html, set a User-Agent |
| **SEC EDGAR XBRL** | Validation layer: audited revenue/income/assets for public filers | No key; User-Agent header required |
| **Damodaran (NYU Stern)** | Industry benchmark margins/ROC for context | Direct .xls; taxonomy mapping needed |

## FLAGSHIP VISUALS

- Size-vs-margin scatter
- Sector profitability boxplots
- Revenue-per-employee ranked bars
- Concentration & HHI trends
- US-state HQ choropleth
- Churn stacked bars + rank bump chart

## METHODOLOGY

1. Typed config with **pinned SHAs/versions**; ingest and cache all raw sources.
2. Land raw verbatim in DuckDB with source + ingestion lineage.
3. Normalize names, types, **units to USD millions**, sector labels, HQ fields.
4. **Cross-year entity resolution** → stable company_id via alias table.
5. Marts: fact_company_year + dim_company; window functions for churn & rank change.
6. Quality gates → 8 analyses → Streamlit + pytest + CI + data dictionary.

## RISKS & GOTCHAS

- Fortune rankings are **proprietary editorial data** — community CSVs are reproductions; cite them as such.
- **URLs drift/404** — pin commit SHA / dataset version and mirror raw files into the repo.
- Employees/sector/HQ columns are **inconsistent across years**; fiscal conventions won't tie exactly to SEC.
- Nominal dollars — **deflate with CPI** before making trend claims.

PROJECT 04 · ADVANCED CAPSTONE

# Cloud, ERP & AI Vendor Intelligence

Segment revenue, implied market position, and security-risk exposure — fusing SEC filings, developer-adoption signals, and the NVD/CISA vulnerability record across ~10 enterprise-tech vendors.

| DIFFICULTY | EFFORT | DATA CLEANLINESS | UNIVERSE | WINDOW |
|---|---|---|---|---|
| **Advanced** | **~60–90 h** | ★★★☆☆ | **~10 vendors** | **FY2016 → now** |
| Highest ceiling, most differentiated | CPE attribution is the hard part | Entity resolution across 3 ID systems | MSFT, AMZN, GOOGL, ORCL, CRM, SAP, NOW, SNOW (+IBM, WDAY) | Filings + 10k+ CVEs + KEV catalog |

## THE INTERVIEW STORY

> "The enterprise-tech vendors compounding revenue fastest are also accumulating the **most severe, most-exploited vulnerabilities** — here's that trade-off, measured, with every number traceable to a primary-source filing or a government vulnerability record."

## THREE PILLARS + A CAPSTONE HYPOTHESIS

**A · Financials**
Reportable-segment revenue, operating income, YoY growth from SEC EDGAR XBRL.

**B · Market position**
Revenue mix within the peer set + free adoption signals — *never* fabricated market share.

**C · Security exposure**
NVD CVE burden, CVSS mix, CISA KEV exploited-rate, publication-to-KEV lag.

**Capstone:** join A × C into a vendor scorecard and test one crisp hypothesis — is higher cloud-revenue growth associated (Spearman) with a heavier severe/exploited vulnerability load?

## KEY ANALYTICAL QUESTIONS

1. How does **AWS's clean segment disclosure** compare to MSFT's blended "Intelligent Cloud" and GCP's line — where does analysis hit the disclosure wall?
2. Do **developer-adoption signals** (Stack Overflow survey, GH Archive) corroborate the revenue momentum ranking?
3. Which vendors carry the largest **CVE burden**, and how has their severity mix shifted YoY?
4. What fraction of each vendor's CVEs became **actively exploited** (CISA KEV) — and how fast?
5. Is faster cloud growth **statistically associated** with heavier severe/exploited load?

# Vendor Intelligence — pipeline, data, risks

## VERIFIED DATA SOURCES

| SOURCE | ROLE | ACCESS |
|---|---|---|
| **NVD CVE API 2.0** | **Primary security dataset** — CVEs per vendor via CPE matching; CVSS severity + dates | Free key → 50 req/30s; paginate, cache |
| **CISA KEV catalog** | Actively-exploited flags → exploited-rate, publication-to-KEV lag | JSON/CSV, no key; CC0 mirror |
| **SEC EDGAR XBRL** | **Primary financial dataset** — total & segment revenue/income per vendor | ⚠ Segment tags non-standard — may parse 10-K notes |
| **Stack Overflow survey** | Adoption corroboration: YoY tool-usage share | ⚠ Self-selected; columns change yearly — directional only |
| **GH Archive / yfinance / MITRE** | Optional: mindshare proxy · equity panel · bulk CVE fallback | BigQuery free tier · cache · shallow clone |

## FLAGSHIP VISUALS

- Segment revenue panel w/ is_disclosed shading
- Revenue-mix treemap
- Adoption-vs-revenue momentum
- CVE burden + severity stacked area
- Exploited-rate ranked bar
- Capstone scorecard scatter

## METHODOLOGY

1. **Vendor crosswalk: ticker ↔ CIK ↔ CPE strings** — the hard entity-resolution artifact.
2. Rate-limited, cached ingest: NVD, KEV, EDGAR segment facts, survey CSVs.
3. **Pillar A** — segment panel + revenue mix; flag undisclosed cloud lines.
4. **Pillar C** — CVE attribution via CPE; KEV join; exploited-rate + lag.
5. **Capstone** — vendor scorecard; Spearman: growth vs. severe/exploited load.
6. pandera gates → Streamlit scorecard → pytest + CI → licensing/limitations README.

## RISKS & HONESTY CONSTRAINTS

- **Never fabricate Azure/GCP dollar revenue** — MSFT/Google disclose growth % only; model the gap with an is_disclosed flag.
- **CPE attribution is imperfect** — counts are estimates; validate a sample, report attribution error.
- **True market share is paywalled** (Gartner/Canalys/Synergy) — press headlines are context only, never a dataset.
- Keep to ~10 vendors — expanding the crosswalk balloons entity-resolution effort; equity analysis stays non-causal.

MASTER DATASET REFERENCE

# The data foundation — free & adversarially verified

Every source behind the four projects passed **43 adversarial verification checks**. No paid data, no ToS-violating scrapes — and the failures found during verification are documented with working replacements.

### ✅ FULLY OPEN — NO ACCOUNT

- **SEC EDGAR XBRL** — 3 of 4 projects; User-Agent header, 10 req/s, US filers only.
- **NVD CVE API 2.0 · CISA KEV · MITRE cvelistV5** — the security record.
- **OWID COVID · Oxford OxCGRT** — cases & stringency (OWID frozen Aug 2024).
- **Damodaran NYU** — industry benchmarks (URLs rotate each January — pin).
- **yfinance** — all projects; fragile (429s) → fetch once, cache, retry.
- **Wikipedia · Stack Overflow survey · World Bank · Bain/Altagamma headlines · company URD PDFs · GH Archive**.

### 🔑 FREE WITH AN INSTANT KEY

- **FRED** — ~120 req/min; macro + VIX series.
- **BEA** — GDP-by-industry (optional).
- **BLS** — 500 queries/day (most series also on FRED).
- **Kaggle** — Fortune enrichment; pin slug + version.

### 🚫 PAYWALLED — EXCLUDED, DO NOT SCRAPE

- Statista · Macrotrends (ToS forbids bots) · stockanalysis.com.
- Gartner / Canalys / Synergy / IDC share reports → replaced by filings, free headlines, and revenue-mix / adoption proxies.

### ⚠️ CORRECTIONS FOUND IN VERIFICATION

- **Stooq direct CSV → blocked** by a JS anti-bot wall. Fix: yfinance / pandas-datareader readers, cached.
- **Fortune combined CSV → 404.** Fix: per-year files 2005–2019, concatenated, SHA-pinned.
- **Financial Modeling Prep → endpoints changed/capped.** Fix: SEC EDGAR XBRL as ground truth.

### THE REUSABLE TOOLKIT

- **SEC EDGAR XBRL client** — used by 3 of 4 projects.
- **yfinance fetcher** with Parquet caching + retry — all four.
- **Damodaran .xls benchmark loader** — Luxury + Fortune 500.
- **Generic cached HTTP client** (retry/back-off, on-disk cache) — every REST/CSV source.

WHY THIS PORTFOLIO WORKS

# Defensible numbers, shared spine, compounding effort

## ONE ENGINEERING SPINE ACROSS ALL FOUR PROJECTS

**Free verified APIs**
EDGAR, FRED, NVD/KEV, OWID, Damodaran…

→

**Cached ingestion**
Fetch once → Parquet; retry/back-off; pinned versions

→

**DuckDB warehouse**
raw → staging → marts, with lineage

→

**Quality gates**
pandera checks + pytest over every formula

→

**Delivery**
Streamlit dashboard + GitHub Actions CI + findings README

## HONESTY AS A FEATURE

- Metrics are **defined where they're defensible** — "market share" means share within the covered, listed peer set, reproducible from primary filings.
- **Disclosure gaps are modeled, not papered over** — is_disclosed flags for Azure/GCP; attribution error reported for CVE↔vendor matching.
- **Crosswalks and caveats are first-class artifacts** — taxonomy mappings, alias tables, FX conventions, and frequency-alignment choices are all documented and tested.
- Each brief ships with **pre-written honest CV bullets** and clearly scoped extensions (backtests, event studies, factor decompositions, auto-refresh).

## THE NARRATIVE ARC

- **Start: Fortune 500** — prove the data-engineering craft and build the shared toolkit (~40–70 h).
- **Flagship: Sector Rotation** — a formal event study with CARs and t-tests; the strongest quant story per hour invested.
- **Recruiter magnet: Luxury Comps** — sell-side comps work: normalization, crosswalks, valuation vs. fundamentals.
- **Capstone: Vendor Intel** — finance × cybersecurity; a measured growth-vs-risk trade-off nobody else in the stack of CVs will have.